

Impact of diet on irinotecan toxicity in mice

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ABSTRACT

Irinotecan is highly effective in the treatment of metastatic colorectal cancer as well as many other cancers. However, irinotecan is known to cause severe diarrhea, which pose significant problems in patients undergoing irinotecan based chemotherapy. Dietary and herbal components have shown promise in improving gastrointestinal health. Therefore, we compared the effect of grain-based chow diet containing phytoestrogens and corn/alfalfa as fat source to purified diets containing either animal-derived fat source (lard) or plant-derived fat source (soybean oil) on irinotecan-induced toxicities in mice. The concentration of the toxic metabolite, SN-38, was measured in the serum, and the activity of main enzyme, carboxylesterase (CEs) involved in biotransformation of irinotecan to SN-38 formation was measured in the liver. We found that the grain-based diet was protective against irinotecan-induced diarrhea. Interestingly, purified diet containing lard caused fatty liver in mice, while grain-based chow diet containing corn/alfa-alfa or purified diet with soybean oil did not cause fat deposition in the liver. Serum SN-38 concentration was significantly higher in the mice fed with purified diets compared to the chow-fed mice. Hepatic CEs activity was induced in the presence of irinotecan in mice on purified diets, but not chow diet. These results indicate that components of grain-based natural diet (presumably phytoestrogens and/or the macronutrients balance) compared to purified diets may have a beneficial effect by controlling the adverse effects of irinotecan in cancer patients.

1. Introduction

The topoisomerase inhibitor, irinotecan (CPT-11, Camptosar[®]), is approved by the FDA to treat metastatic colorectal cancer and is also used in many other cancers. Although irinotecan is highly effective, its use is severely limited due to its adverse effects [1]. Irinotecan causes diarrhea in ~88% patients, and up to 40% patients experience severe late onset (grade 3–4) diarrhea which can be life threatening [2–6]. Irinotecan-based chemotherapy also causes non-alcoholic fatty liver disease (NAFLD), that worsens postoperative outcome like morbidity and mortality in colorectal liver metastases (CLM) patients [7,8].

Understanding of irinotecan's complex pharmacology and metabolism has provided clues to the etiology of irinotecan-induced diarrhea. Irinotecan is a prodrug that is converted to the active form, 7-ethyl-10-hydroxycamptothecin (SN-38), by carboxylesterases (CEs). SN-38 is further metabolized by glucuronosyltransferases (UGTs) to inactive SN-

38 glucuronide (SN-38G) [9], which can be deconjugated back to SN-38 by bacterial β -glucuronidases in the gut. Both the parent compound and metabolites undergo enterohepatic circulation and accumulation of toxic SN-38 in blood and intestine due to this process has been correlated with diarrhea in humans [3,10,11].

Significant effort has been made to reduce severity of diarrhea with irinotecan. Different strategies have been tested in preclinical models and clinical studies including dose modification, intestinal alkalization, structural modification, pharmacological therapies, drug metabolizing enzymes and transporter inhibitors, probiotics, antibiotics, and other miscellaneous agents [12–18]. Currently, loperamide, an anti-diarrheal drug, is being used to control diarrhea associated with irinotecan [19,20]. Although, high dose of loperamide is moderately effective in uncomplicated diarrhea, its utility as monotherapy for severe diarrhea is limited [21]. Therefore, new approaches to prevent intestinal toxicity of irinotecan-based regimens remains an on-going effort. At present,

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Table 1
Major components in the diets used in the experiment [40,44–46].

Compositions		Chow	PD-1	PD-2
Type		Grain/cereal based diet	Purified diets	
Protein	Kcal (%)	20	20.	20
	Source	Multiple ^a	Casein, Lactic, 30 Mesh	
Carbohydrate	Kcal (%)	64	64	70
	Source	Soya bean, plant	Corn starch, sucrose	
	Sucrose (%)	0	8	35
	Fiber	High (4X) fiber content: cellulose, dextrins, inulin, lignin, waxes, chitins, pectins, beta-glucans and oligosaccharides	Low fiber content 5% cellulose	
Fat	KCal %	13	16	10
	Source	Ground wheat/corn/oats/alfalfa (Veg)	Soya bean oil (Veg)	
	Saturated (g/100 g)	15	16	39
	Monosaturated (g/100 g)	28	23	45
	Polyunsaturated (g/100 g)	55–68	58	11
phytoestrogens	–	^b Isoflavones (398 Aglycone Units, AU)	Not detected	

^a Complete ingredient contents that are source of protein and others can be found in the diet specific sheets.

^b Data on phytoestrogen content was provided by Lab diet (personal communication).

there are no interventions to counteract the NAFLD caused by irinotecan, and preoperative assessment for significant liver pathology with the currently available tools is a challenging clinical problem. Therefore, strategies need to be developed to reduce/prevent irinotecan gut and liver toxicity to improve therapeutic outcome in patients.

Specific dietary elements have attracted attention towards improving the therapeutic outcomes of chemotherapeutic agents by reducing toxicities. In this context, n-3 polyunsaturated fatty acids (PUFA) eicosapentanoic acid, and prebiotic oligosaccharides have been proposed to be potential modulators of gut injury related to cancer chemotherapy [22–24]. Significant interest has been given towards prebiotics for managing diarrheas involving disruption of intestinal flora balance. Also, dietary monounsaturated fatty acids (MUFAs) when replaced with dietary carbohydrate and saturated fat, have shown to prevent the development of NAFLD [25]. Several studies also support the protective effects of n-3 polyunsaturated fatty acids (PUFAs) in NAFLD [26,27]. Natural phenols, including resveratrol and anthocyanins have shown positive effects on animal NAFLD models [28], however their benefits in patients with NAFLD either have failed to show any significantly improvements [29] or has not yet been well elucidated [30].

Dietary compounds improve the efficiency of cytotoxic agents [31] and alleviate adverse effects [31–33]. Therefore, nutritional approach to reduce adverse side effects and improve efficacy of irinotecan by reducing its dose-limiting intestinal toxicity has been of prime interest recently. Dietary supplementation with the non-essential amino acids, glutamine showed improvement in irinotecan-induced grade 4 diarrhea in small study series with five patients [34]. In rats, glutamine treatment reduced the incidence of severe diarrhea induced by irinotecan [35]. Likewise, reduced apoptosis in the crypts of the small intestine with irinotecan was observed in mice fed with 3% AAFA, a omega-3 polyunsaturated fatty acid product containing a high concentration of long chain fatty acids [36,37]. In addition, these mice were also protected against irinotecan-induced liver hypertrophy (enlargement of the liver) [36]. However, optimal strategies to translate these findings into clinical care remain to be elucidated.

In our current study, we tested the influence of grain-based and purified diets on irinotecan associated toxicities. Our study revealed a diet-dependent toxic effect of irinotecan, as the grain based chow-fed mice were protected from irinotecan-induced toxicities, whereas mice on purified diets that are non-grain based developed toxicities with irinotecan.

2. Materials & methods

2.1. Chemicals

Camptothecin (CPT; internal standard (I.S)), 7-ethyl-10-hydroxycamptothecin (SN-38), p-nitrophenyl-β-D-glucuronide (Cat # N1627), and 4-Nitrophenyl acetate (PNPA; Cat # N8130) were purchased from Sigma-Aldrich (St. Louis, MO). Irinotecan hydrochloride (NDC-0703-4437-11) for injection was procured from Martin surgical (Houston, TX). SN-38G was prepared in our lab using cell lines we previously reported [38]. Acetonitrile and waters (LC-MS grade) were purchased for VWR international, LLC (Suwanee, GA, USA). Unless specified, all other materials were purchased from Sigma-Aldrich (St Louis, MO, USA.).

2.2. Animals and diets

The C57BL/6J mice (male, 12-weeks old) were purchased from Jackson Laboratory (Bar Harbor, ME). Three different diets were used in the current study, primarily categorized as grain/cereal based and purified ingredients based; (a) Chow diet (#0007688); (b) American Institute of Nutrition modified (AIN) 93G (# D10012G); and (c) Low fat diet (# D12450B). Chow diet was purchased from Pico Lab and purified diet from Research Diets, Inc. For understanding the basic differences among the diets used, only the major components of these diets are listed in Table 1 but the breakdown of ingredients for each of the diet can be found in product specific sheets. Chow diets are grain or cereal based and typically contain ingredients such as ground corn, ground oats, alfalfa meal, soybean meal and ground wheat. Chow diets are 'closed' formulas, thus exact amount of each ingredient is mostly unknown [35,39]. In comparison to chow diet, the purified diets are considered open formulas and have well-characterized definition of the lipid, protein, and carbohydrate constituents. American Institute of Nutrition (AIN) formulation, AIN-76 A is commercial source for semi-purified diets and AIN-93G used in our study is redefined AIN-76 A formulation [40] and is referred throughout the manuscript as purified diet 1 (PD-1). Low-fat diet is control diet used as matched controls for purified high-fat diet [41] and has defined, and flexible formulation similar to AIN-93G. However, this low-fat purified diet predominantly differs from AIN-93G in lard being the source of fat instead of soybean oil (Table 1). Throughout the manuscript, this low-fat diet is referred as purified diet 2 (PD-2). A major difference between grain-based chow diet and purified diets is that, they differ greatly in their phytoestrogen content. Though the exact amount of phytoestrogen is unknown for chow diet, this grain-based diet is primarily made with ground plants

and therefore contains abundant phytoestrogen as a source of protein [42]. On the other hand, the purified diets contain casein as the protein source, which is known to be free of phytoestrogens. Also, both soybean oil and lard as the sources of fat, lacks isoflavones (class of phytoestrogens) because these are not fat soluble [42,43].

The animals were fed with these different diets for 3 weeks, *ad libitum*. All the animal care and use protocols were approved by the Institutional Animal Care and Use Committee guidelines. Animals were maintained in 12 h dark/light cycle and temperature-and-humidity-controlled environment.

2.3. Irinotecan treatments

For diet effect study, mice ($n = 5-8$) were administered intraperitoneally (i.p.) with vehicle or irinotecan hydrochloride solution (50 mg/kg) for 6 consecutive days. On day 7, following euthanasia, livers and blood were collected 24 h after the last injection.

2.4. Diarrhea and body weight evaluation

The severity of the diarrhea and body weight were monitored every day. Diarrhea was scored using a scale described earlier [47] with slight modification: 0 (normal; normal stool or absent); 1 (slight; slightly wet and soft stool); 2 (moderate, wet and unformed stool with moderate perianal staining of the coat); 3 (severe, watery stool with severe perianal staining of the coat); and > 3 (life threatening, bloody stool with severe perianal staining of the coat).

2.5. Histological analysis

Liver sections ($n = 4$ to 6 for each treatment group) were fixed in 10% formalin, embedded in paraffin, and stained with hematoxylin and eosin (H & E) at Human Tissue Acquisition and Pathology, Baylor College of Medicine, Texas. Morphological findings for liver were recorded in a semi-quantitative manner [48,49]. Estimation of steatosis percentage is routine in the interpretation of liver biopsies and is an important part of the diagnosis for steatohepatitis (critical stage of NAFLD). Steatosis was graded as none (score 0) minimal (score < 1 ; $< 5\%$), mild (score $1 +$; $5-30\%$), moderate (score $2 +$; $30-60\%$), or severe (score $3 +$ to 4 ; $> 66\%$) based on percentage of hepatocytes showing fat accumulation in the sections (5 fields per $5 \times$ slide, concentrated around perivenular regions/acinar zone 3). Each slide was scored individually and average score of each group was considered for reporting as final readout. All histological slides reviewed by an experienced pathologist (MI) at Baylor College of Medicine, Texas.

The standard experimental method was used to gather unbiased data on liver damage. The pathologist had no knowledge of specific details on experimental groups and were provided with only numbered slides to mask information on groups and treatment. Pathologist carefully observed any change in the tissue that deviate from “normal” and characterized the nature based on visual memory and basic principles learned in special pathology and has reinforced by years of practice.

2.6. LC-MS/MS sample preparation and analysis

The analytes, CPT, irinotecan, SN-38 and SN-38G were quantified using validated LC-MS method published previously with few modifications [50,51]. Briefly, the serum samples were extracted with acidified extraction solvent (acetonitrile: methanol = 1:1 acidified with 1% glacial acetic acid) containing $0.5 \mu\text{g/ml}$ of CPT (I.S.). After centrifugation, the supernatant was collected, and the solvent was removed under an air flow. The residue was reconstituted in 50%ACN/50% water/1% GAA by vortex-mixing for 30 s for LC-MS injection after centrifugation ($15,000 \times g$, 15 min, 4°C).

The LC conditions were: Waters Acquity™ UPLC system. Analysis was carried out using Acquity UPLC BEH C_{18} Column

($2.1 \text{ mm} \times 50 \text{ mm}$, 300°A , 1.7 μm , Waters, Milford, MA, USA). Mobile phase A (0.1% formic acid in water) and mobile phase B (100% acetonitrile) were operated with a gradient elution at a flow rate of 0.4 ml/min as follows: 10% B $\rightarrow$ 25% B (0–0.5 min), 25% B $\rightarrow$ 40% B (0.5–1 min), 40% B (1–2.5 min), 40% B $\rightarrow$ 10% B (2.5–4.5 min). The column temperature and sample temperature were 60 or 4°C . The injection volume was $10 \mu\text{L}$. The MS conditions were: API 5500 Qtrap triple quadrupole mass spectrophotometer (AB Sciex, USA) equipped with a Turbospray™ source as described in our previous publication. The system was operated in positive electrospray ionization (ESI) and multiple reactions monitoring (MRM) scan mode.

2.7. Carboxylesterase activity

Whole cell extracts were prepared as described previously [51–53]. Briefly, the livers were homogenized in 1 ml of homogenization buffer and were centrifuged at 9000 rpm for 15 min at 4°C . The supernatants were collected and stored in aliquots in -80°C . The protein concentrations were determined by bicinchoninic acid (BCA) assay according to the manufacturer's protocol (Pierce, Rockford, IL, U.S.A). Total carboxylesterase (CEs) activity as formation of p-nitrophenolate was monitored over 30 min at 405 nm [53] by a spectrophotometric method described previously [54]. The prototypical substrate PNPA was used at $750 \mu\text{M}$ and the specific activity is expressed as $\mu\text{moles PNP formed mg protein}^{-1} \text{ min}^{-1}$.

2.8. β -glucuronidase assay

β -glucuronidase activity in fecal samples was determined by colorimetric assay as described previously by using an absorbance spectrophotometer (DU800; Beckman Coulter Inc., Indianapolis, IN) [55,56]. To the fecal suspension, p-nitrophenyl- β -D-glucuronide was added as substrate and incubated at 37°C water bath. The supernatant collected at end of the reaction were measured for absorption at 405 nm [53].

2.9. Statistical analysis

All data are presented as mean $\pm$ S.D. from 4 to 5 mice per group. Differences between treatment groups were assessed for statistical significance by un-paired Student's t-test, or two-way ANOVA followed by the Bonferroni post-hoc test where appropriate, using GraphPad Prism 4.0 (GraphPad Software, Inc., La Jolla, CA, USA). Statistical significance was demonstrated at $p < 0.05$.

3. Results

3.1. Effect of diets on irinotecan-induced diarrhea

The diarrhea and body weight were recorded to evaluate the effect of diets on irinotecan-induced gastrointestinal (GI) toxicity. All mice on different diets with no irinotecan treatment, maintained weight (Fig. 1, vehicle group) and did not develop diarrhea (0% incidences) or liver toxicity throughout the experiment duration. With irinotecan treatment, mice on chow diet didn't develop diarrhea (Table 2) and maintained constant body weight with initial weight loss for two days (Fig. 1). In contrast, mice fed with purified diets, PD-1 and PD-2, when treated with irinotecan, developed diarrhea by day 4 of irinotecan injections, which worsened to bloody diarrhea by day 7 (Table 2). This was accompanied by weight loss in irinotecan-treated mice fed with these purified diets.

3.2. Effect of diet on irinotecan-induced steatosis

As shown in Fig. 2, mice fed with purified diet containing lard, PD-2, developed severe hepatic micro- and macrovesicular steatosis,

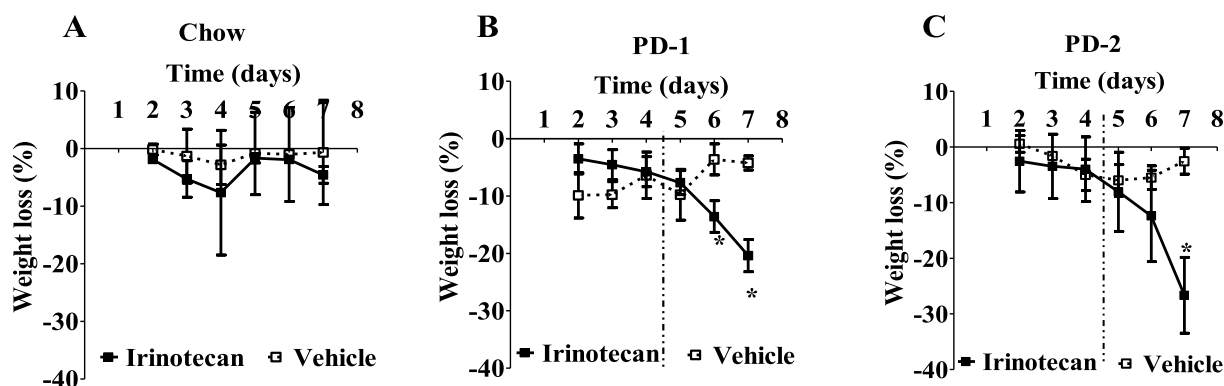


Fig. 1. Body weights of vehicle and irinotecan treated mice. Values are means $\pm$ SD, $n = 5$ for chow and PD-1 diet group and $n = 8$ mice for PD-2 diet group. (|) indicates the day, incidence of either loose, watery or bloody stool appeared in mice. Data shows that chow diet protects mice from weight loss associated with irinotecan-induced diarrhea. $*p < 0.05$ compared to vehicle.

Table 2

Incidences of diarrhea. The values are expressed as the mean diarrhea score; % mice that developed 1–3 grade diarrhea.

Irinotecan, 50 mg/kg/ day	# of mice	Incidence of diarrhea Score (0 - > 3); % of mice with diarrhea (0–25)						
		Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Chow	4	0; 0%	0; 0%	0; 0%	0; 0%	0; 0%	0; 0%	0; 0%
PD-1	4	0; 0%	0; 0%	0; 0%	1; 25%	1; 25%	> 3; 75%	> 3; 100%
PD-2	8	0; 0%	0; 0%	0; 0%	1; 25%	2; 50%	3; 65%	> 3; 88%

characterized as fat accumulation in liver, with mean score of 2 in 75% of mice treated with irinotecan (Fig. 2F). Mice fed with chow diet or PD-1 (containing soybean oil as the fat source) had no fat deposition in the liver. As we detected irinotecan-induced hepatic steatosis only in PD-2 fed mice, it is reasonable to assume that there exists a relationship between the irinotecan-induced fatty liver and the dietary intake of lard, which is mainly long-chain, saturated fatty acid (LC-SFA) and monounsaturated fatty acids (MUFA) (Table 1).

A striking finding was that in the group of mice that developed hepatic fat accumulation with irinotecan, the extent of steatosis largely varied among mice within the same treatment group, ranging from

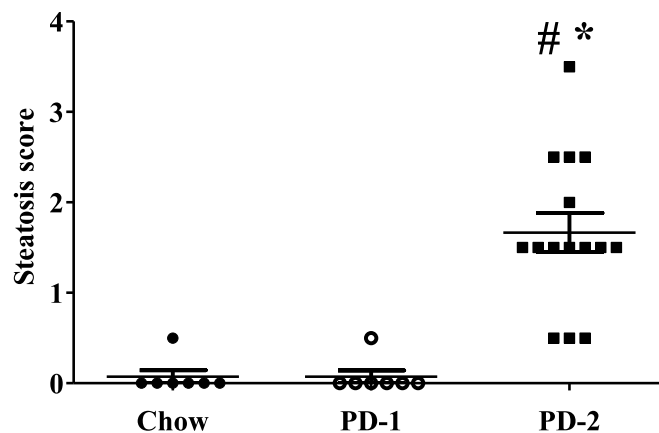


Fig. 3. The graph plots the steatosis scores of H&E stained liver sections from irinotecan treated mice on different diets. Mean steatosis scores in PD-2 fed mice are significantly higher than chow and PD-1 fed mice ($*p < 0.05$ compared to chow & $\# p < 0.05$ compared to PD-1, one-way ANOVA). ● ($n = 7$, chow), ○ ($n = 7$, PD-1) and ■ ($n = 17$, PD-2), each symbol represents steatosis score for individual animal.

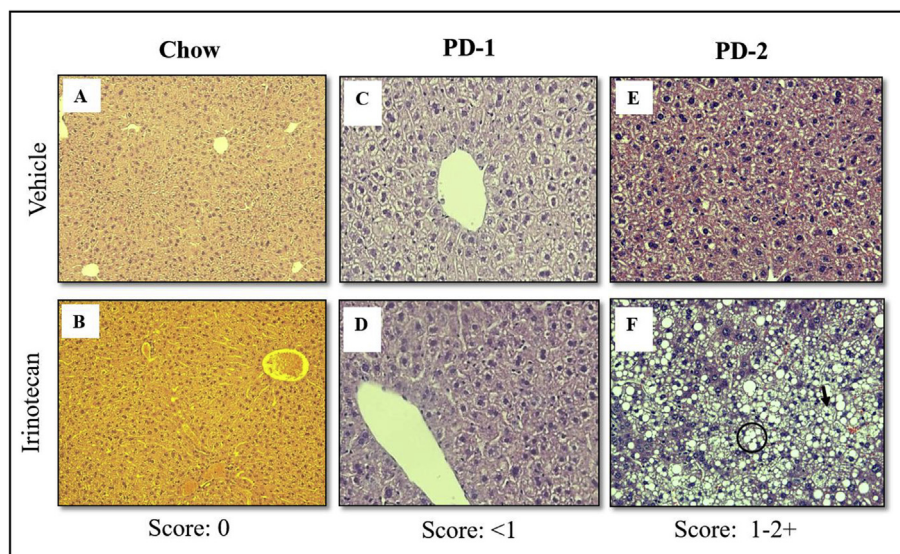


Fig. 2. Hematoxylin and eosin (H&E)-stained sections of representative liver sections from irinotecan or vehicle treated C57BL/6 mice fed on different diets. Hepatic steatosis is absent in mice fed on Chow diet (A–B) and PD-1 (C–D). Irinotecan-induced mixed hepatic steatosis (micro & macro vesicular) in mice maintained on PD-2 (E–F). 200 $\times$ magnification. Accumulated fats indicate microvesicular as (←) and macro as (○).

Serum levels

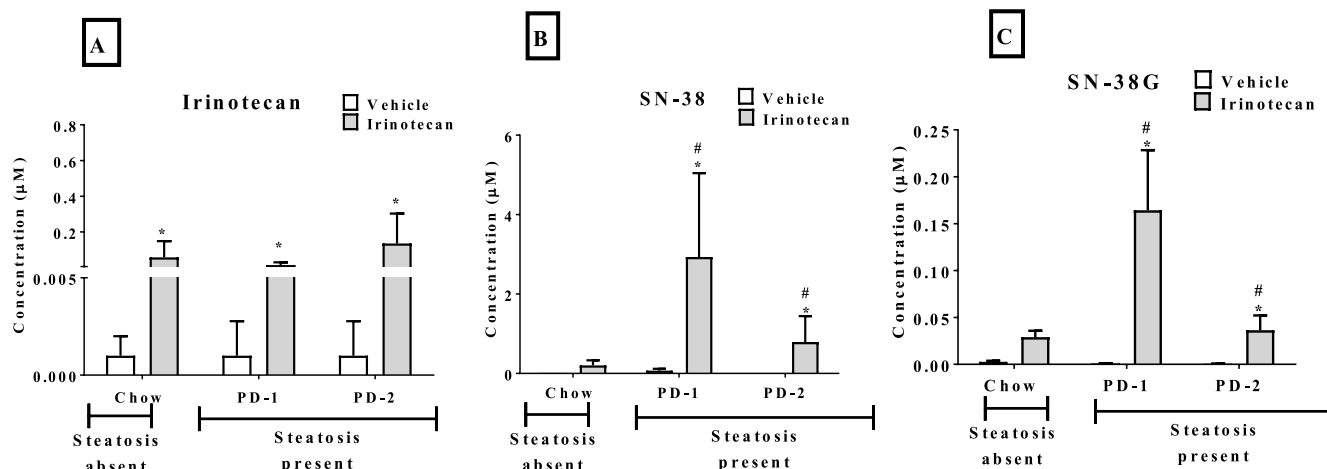


Fig. 4. Concentrations of irinotecan and its metabolites at 24hr of last dosing are increased in mice fed with purified diet compared to chow diet. Values are means $\pm$ S. D, n = 5 for chow and PD-1 diet group and n = 8 mice for PD-2 diet group. (*) p < 0.05 compared to respective vehicle group and (#) p < 0.05 compared to irinotecan treated chow group.

none to micro and macro-vesicular hepatic steatosis (Fig. 2). Some mice developed severe macrovesicular steatosis and others had centrilobular microvesicular steatosis. Overall, in an average, > 75% of animals, had massive steatosis in liver. Fig. 3 shows a plot of histological examination based steatosis score of individual mouse after i. p injection of 50 mg/kg irinotecan for 5 days, demonstrating a significant increase in scores with PD-2 diet.

3.3. Effect of diet on the serum concentration of irinotecan and its metabolites

We determined the effect of diet on serum levels of irinotecan and its metabolite. On day 7, after 24hr of 6th day treatment, mice were euthanized, and blood were collected to measure the serum concentrations of irinotecan and its metabolites.

The mean concentrations at 24hr time point of irinotecan were similar for all diets (Fig. 4A). The mean concentration of SN-38 in both purified diets (PD-1 & PD-2) were significantly higher when compared to chow diet (Fig. 4 B). The SN-38 concentration was similar between the two purified diets, although its level appears to be higher in PD-1 fed mice mostly because of large standard deviation (S.D). Similar trend was observed for mean concentration of SN-38G as well (Fig. 4C). The large S. D in PD-1 fed group could be possibly due to greater variability in occurrence of diarrhea (Table 2).

3.4. Effect of diet on CEs and β -glucuronidase activities

Carboxylesterases (CEs) are the main enzymes which convert irinotecan to SN-38. So, we measured hepatic CEs activity in the mice fed with chow or purified diets in the presence or absence of irinotecan. We did not find any changes in the basal activity of these enzymes in the mice on the three different diets (Fig. 5A). However, the activity of the CEs enzymes was significantly higher in the presence of irinotecan compared to the vehicle in mice fed with purified diets, but not in the chow-fed mice.

Among several mechanisms for the development of irinotecan-induced diarrhea, increases in intestinal β -glucuronidase levels caused by changes in intestinal bacteria has also been proposed [57,58]. In our current study, to begin with, the basal fecal β -glucuronidase activity was higher in purified diet (PD1 and PD2) fed mice compared to chow diet fed mice. Although irinotecan treatment decreased the fecal β -glucuronidase activity in all groups, the enzyme activity was relatively

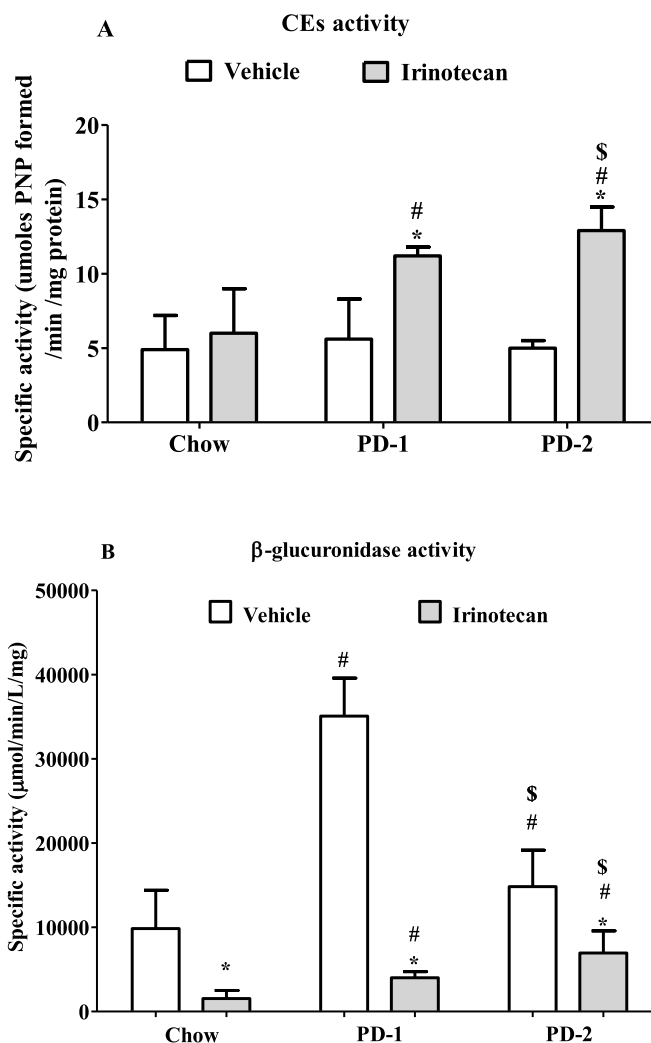


Fig. 5. CEs and β -glucuronidase activities were altered in irinotecan treated mice fed with different diets. Values are means $\pm$ S. D, n = 5 for chow and PD-1 diet group and n = 8 mice for PD-2 diet group. (*) p < 0.05 compared to vehicle group, (#) p < 0.05 compared to chow, (\$) p < 0.05 compared to PD-1.

higher in purified diets compared to chow diet (Fig. 5B), indicating possible mechanism leading to higher levels of serum SN-38 in PD-1 & PD-2 diet fed mice (Fig. 4).

4. Discussion

The objective of this study was to understand effect of diet on irinotecan-induced gut and liver toxicity. We found that the grain-based chow diet was protective against irinotecan toxicity, whereas PD diets (lacking phytoestrogens and with different composition of macronutrients) accentuated diarrhea, and additionally the lard-containing PD-2 also led to steatosis in irinotecan-treated mice. There are significant differences in the composition of the chow diet and the purified diets, and one or more ingredients of the chow diet could have prevented toxicities associated with irinotecan by multiple pathways. The chow is normally low in fat (3% of energy), high in carbohydrates (approximately 50% starch), rich in phytoestrogen, 28% protein and no lard [35,59]. Dietary phytoestrogens have been reported to beneficial through targeting mechanism involved in NAFLD, including decrease lipogenesis and lipolysis in adipocytes and reduce fat deposition [60,61]. Furthermore, as fat source in chow diet is vegetable based, this diet is potentially high in n-3 polyunsaturated fatty acid (PUFA) (Table 1) [45], and several studies have shown that diets enriched with PUFA steatohepatitis (advanced stage of NAFLD) in rats [62,63]. Therefore, PUFA and phytoestrogens can contribute to the protective effect of chow diet. In a separate study, we found that irinotecan-induced steatosis was associated with disrupted fat homeostasis and hepatic inflammation (data not shown). Also, the chow is a high fiber diet containing complex carbohydrates, with fats from a variety of vegetable sources. Consumption of dietary fibers reduces the risk of colorectal cancer (CRC) by maintaining a healthy gut microbiota population [64–66]. Therefore, it is possible that in addition to PUFA and phytoestrogen, large content of fiber in chow diet may contribute towards protecting the mice from developing intestinal toxicity after irinotecan treatment.

Purified diets, PD-1 and PD-2 mostly differ in their source of fat since vegetable-based soybean oil is the source of fat in PD-1, whereas lard contribute to the fat content in PD-2. In our study, mice fed lard containing PD-2 diet developed steatosis with irinotecan, whereas, mice on soybean-containing PD-1 diet had no hepatic fat accumulation with irinotecan treatment (Fig. 2). Lard-based high-fat diets are associated with increased body weight, liver triglycerides (TGs) deposition, increased concentrations of plasma TGs, free fatty acids (FFAs), and insulin [67,68]. It is well-known that lard and sucrose have damaging effect towards the development and severity of NAFLD [69,70]. Lard is an abundant source of saturated fatty acids (39.2%) and a poor source of PUFA (11.2%) compared to soybean oil, which is especially rich in PUFA [41]. In contrast to lard, soybean oil has very low saturated fat and high PUFA (Table 1). Therefore, the attenuation of irinotecan-induced fat accumulation observed with PD-1 diet suggests balanced fatty acid pool and normal insulin sensitivity that possibly allowed maintenance of fat homeostasis. However, PD-1 diet did not protect mice from irinotecan-induced diarrhea (Table 2). The sole source of fiber in PD-1 is cellulose and unlike other fibers, cellulose has little if any growth-promoting activity for intestinal microbes [71,72]. It can therefore be assumed that soluble fiber content in PD-1 was not effective enough to prevent intestinal toxicity with irinotecan.

Drug levels and its effects are unpredictable and do not correlate well with the type of liver injury, its severity, or liver function test results. Thus, no general rules are available for modifying drug dosage in patients with liver disease. We collected blood at 24hr after 6th day treatment (last day injection) to estimate concentrations of irinotecan and its metabolites so as to determine any relation between drug levels and irinotecan toxicities. In our study, we observed similar serum irinotecan levels in mice irrespective of which diet they were fed, however the serum levels of both metabolites, SN-38 and SN-38G were

significantly higher in mice fed with purified diets compared to chow diet (Fig. 4). Although, high concentrations of SN-38 is known to cause diarrhea, these studies indicate for the first time that accumulation of serum SN-38 may also be associated with fatty liver.

Since the PK of irinotecan primarily depends on the CEs activity [73–75], we studied its activity in mice fed on different diets. In our study, we found that hepatic CEs activity was increased in mice fed with purified diets compared to chow diet (Fig. 5A), in parallel to increased serum SN-38 levels (Fig. 4), suggesting potential cause and effect relationship. Hydrolysis of SN-38G to SN-38 by enteric bacterial β -glucuronidases can also contribute to higher levels of SN-38 and associated toxicity [57,76]. In our previous study we had observed induction in fecal β -glucuronidase enzyme activity in irinotecan-treated diet induced obese mice correlated to increased levels of SN-38 [53]. In rats, protective effects of dietary fiber on irinotecan toxicity were found not to be mediated by reduced activity of intestinal glucuronidase enzyme activity [77]. However, in our current study, the mice on purified diets with higher serum SN-38 levels also had induced fecal β -glucuronidase enzyme activity compared to mice on chow diet (Fig. 5B), indicating this mechanism possibly plays a role in diet-based effect on irinotecan-induced toxicities. Further mechanistic studies will provide better understanding on the intricate mechanisms responsible and pave way to design intervention to prevent irinotecan-induced toxicities. Several mechanisms have been proposed for the development of irinotecan-induced diarrhea. Based on important contribution of UGT1A1 in irinotecan's metabolism and the existence of genetic polymorphisms, its role in irinotecan induced intestinal toxicity has also been implicated [9,78,79]. In our preliminary analysis, UGT gene expression remained unaltered in vehicle treated mice on different (Supplemental Fig. 1). However, the basal UGT1A1 enzyme activity was higher in the livers of mice on PD-2 diet compared to chow and PD-1 diets (Supplemental Fig. 2). Because the UGT1A1 activity was studied in mice which were not treated with irinotecan, these results indicate that diets can affect the basal UGT activity in the liver. Our results in Fig. 4 shows that SN-38G levels were higher in PD-2 diet compared to the chow diet, which is in agreement with increased basal UGT1A1 activity in PD-2 mice, indicating that increased UGT activity might have resulted in increased SN-38G levels in PD-2 diets. This suggests that a comprehensive understanding of the contribution of enzymes and transporters in regulating irinotecan metabolite levels in mice fed with different diets is warranted. As the irinotecan treated groups had SN-38 and SN-38G levels to begin with, UGT activity estimation in these groups will be studied with different UGT substrates. Temporal studies to investigate the effect of diet and irinotecan exposure on UGT1A1 expression and activity in the liver and gut are underway in our laboratory.

Irinotecan has been applied for the treatment of solid tumors for more than 10 years, however its administration was associated with significant incidence of late-onset diarrhea, which has been recognized as the rate-limiting toxicity of this compound [80,81]. Recent studies report that irinotecan chemotherapy is associated with NAFLD that increases risk for mortality and morbidity after hepatic resection in colorectal liver metastatic patients [7,8,82,83]. Based on our data, presence of lard, which is main source of saturated fats and high sucrose content can promote NAFLD with irinotecan treatment. Our results further suggest that presence of fibers, phytoestrogens and high PUFA, are potentially protecting liver and intestine from irinotecan toxicities. Hence, based on such evidence, careful consideration need to be given on the provision of the nutritional components with irinotecan treatment. Modulation of diet by addition of specific plant-based ingredients may improve the therapeutic outcome of cancer patients undergoing irinotecan treatment.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.cbi.2018.06.018>.

References

- [1] H. Bleiberg, E. Cvitkovic, Characterisation and clinical management of CPT-11 (irinotecan)-induced adverse events: the European perspective, *Eur. J. Canc.* 32A (Suppl. 3) (1996) S18–S23.
- [2] L.B. Saltz, et al., Irinotecan plus fluorouracil and leucovorin for metastatic colorectal cancer. Irinotecan Study Group, *N. Engl. J. Med.* 343 (13) (2000) 905–914.
- [3] J.A. Conti, et al., Irinotecan is an active agent in untreated patients with metastatic colorectal cancer, *J. Clin. Oncol.* 14 (3) (1996) 709–715.
- [4] F. Saliba, et al., Pathophysiology and therapy of irinotecan-induced delayed-onset diarrhea in patients with advanced colorectal cancer: a prospective assessment, *J. Clin. Oncol.* 16 (8) (1998) 2745–2751.
- [5] M.L. Rothenberg, et al., Mortality associated with irinotecan plus bolus fluorouracil/leucovorin: summary findings of an independent panel, *J. Clin. Oncol.* 19 (18) (2001) 3801–3807.
- [6] J.Y. Douillard, et al., Irinotecan combined with fluorouracil compared with fluorouracil alone as first-line treatment for metastatic colorectal cancer: a multicentre randomised trial, *Lancet* 355 (9209) (2000) 1041–1047.
- [7] J.N. Vauthey, et al., Chemotherapy regimen predicts steatohepatitis and an increase in 90-day mortality after surgery for hepatic colorectal metastases, *J. Clin. Oncol.* 24 (13) (2006) 2065–2072.
- [8] D. Zorzi, et al., Chemotherapy-associated hepatotoxicity and surgery for colorectal liver metastases, *Br. J. Surg.* 94 (3) (2007) 274–286.
- [9] R. Atsumi, W. Suzuki, H. Hakusui, Identification of the metabolites of irinotecan, a new derivative of camptothecin, in rat bile and its biliary excretion, *Xenobiotica* 21 (9) (1991) 1159–1169.
- [10] S. Kudoh, et al., Relationship between the pharmacokinetics of irinotecan and diarrhea during combination chemotherapy with cisplatin, *Jpn. J. Canc. Res.* 86 (4) (1995) 406–413.
- [11] Y. Sasaki, et al., A pharmacokinetic and pharmacodynamic analysis of CPT-11 and its active metabolite SN-38, *Jpn. J. Canc. Res.* 86 (1) (1995) 101–110.
- [12] D.F. Kehrer, et al., Modulation of irinotecan-induced diarrhea by cotreatment with neomycin in cancer patients, *Clin. Canc. Res.* 7 (5) (2001) 1136–1141.
- [13] D. Flieger, et al., Phase II clinical trial for prevention of delayed diarrhea with cholestyramine/levofloxacin in the second-line treatment with irinotecan biweekly in patients with metastatic colorectal carcinoma, *Oncology* 72 (1–2) (2007) 10–16.
- [14] M. Michael, et al., Phase II study of activated charcoal to prevent irinotecan-induced diarrhea, *J. Clin. Oncol.* 22 (21) (2004) 4410–4417.
- [15] G.C. Sergio, G.M. Felix, J.V. Luis, Activated charcoal to prevent irinotecan-induced diarrhea in children, *Pediatr. Blood Canc.* 51 (1) (2008) 49–52.
- [16] S. Clarke, P. Tobin, L. Noney, et al., A novel approach to reducing diarrhoea with irinotecan, *Progr. Proc. Am. Soc. Clin. Oncol.* 22 (2003) 133.
- [17] Y. Sakata, H. Suzuki, T. Kamataki, Preventive effect of TJ-14, a kampo (Chinese herb) medicine, on diarrhea induced by irinotecan hydrochloride (CPT-11), *Gan To Kagaku Ryoho* 21 (8) (1994) 1241–1244.
- [18] U. Swami, S. Goel, S. Mani, Therapeutic targeting of CPT-11 induced diarrhea: a case for prophylaxis, *Curr. Drug Targets* 14 (7) (2013) 777–797.
- [19] D. Abigeres, et al., Irinotecan (CPT-11) high-dose escalation using intensive high-dose loperamide to control diarrhea, *J. Natl. Cancer Inst.* 86 (6) (1994) 446–449.
- [20] B.H. Lenfers, et al., Substantial activity of budesonide in patients with irinotecan (CPT-11) and 5-fluorouracil induced diarrhea and failure of loperamide treatment, *Ann. Oncol.* 10 (10) (1999) 1251–1253.
- [21] L.B. Saltz, Understanding and managing chemotherapy-induced diarrhea, *J. Support Oncol.* 1 (1) (2003) 35–46 discussion 38–41, 45–6.
- [22] M.A. Funk, D.H. Baker, Effect of fiber, protein source and time of feeding on methotrexate toxicity in rats, *J. Nutr.* 121 (10) (1991) 1673–1683.
- [23] Y. Kimura, H. Okuda, Prevention by chitosan of myelotoxicity, gastrointestinal toxicity and immunocompetent organic toxicity induced by 5-fluorouracil without loss of antitumor activity in mice, *Jpn. J. Canc. Res.* 90 (7) (1999) 765–774.
- [24] A. Macdonald, Omega-3 fatty acids as adjunctive therapy in Crohns disease, *Gastroenterol. Nurs.* 29 (4) (2006) 295–301 quiz 302–3.
- [25] N. Hancu, et al., Metabolic syndrome—practical approach, *Rom. J. Intern. Med.* 42 (2) (2004) 237–245.
- [26] M. Capanni, et al., Prolonged n-3 polyunsaturated fatty acid supplementation ameliorates hepatic steatosis in patients with non-alcoholic fatty liver disease: a pilot study, *Aliment. Pharmacol. Ther.* 23 (8) (2006) 1143–1151.
- [27] N. Tanaka, et al., Highly purified eicosapentaenoic acid treatment improves non-alcoholic steatohepatitis, *J. Clin. Gastroenterol.* 42 (4) (2008) 413–418.
- [28] L. Bujanda, et al., Resveratrol inhibits nonalcoholic fatty liver disease in rats, *BMC Gastroenterol.* 8 (2008) 40.
- [29] V.S. Chachay, et al., Resveratrol does not benefit patients with nonalcoholic fatty liver disease, *Clin. Gastroenterol. Hepatol.* 12 (12) (2014) 2092–2103 e1–6.
- [30] L. Valenti, et al., Dietary anthocyanins as nutritional therapy for nonalcoholic fatty liver disease, *Oxid Med Cell Longev* 2013 (2013) 145421.
- [31] K.N. Prasad, et al., Scientific rationale for using high-dose multiple micronutrients as an adjunct to standard and experimental cancer therapies, *J. Am. Coll. Nutr.* 20 (5 Suppl.) (2001) 450S–463S discussion 473S–475S.
- [32] B. Jung, N. Ahmad, Melatonin in cancer management: progress and promise, *Cancer Res.* 66 (20) (2006) 9789–9793.
- [33] D.W. Lamson, M.S. Brignall, Antioxidants in cancer therapy; their actions and interactions with oncologic therapies, *Alternative Med. Rev.* 4 (5) (1999) 304–329.
- [34] D.M. Savarese, et al., Prevention of chemotherapy and radiation toxicity with glutamine, *Canc. Treat Rev.* 29 (6) (2003) 501–513.
- [35] H. Xue, et al., Nutritional modulation of antitumor efficacy and diarrhea toxicity related to irinotecan chemotherapy in rats bearing the ward colon tumor, *Clin. Canc. Res.* 13 (23) (2007) 7146–7154.
- [36] W.E. Hardman, M.P. Moyer, I.L. Cameron, Consumption of an omega-3 fatty acids product, INCELL AFAA, reduced side-effects of CPT-11 (irinotecan) in mice, *Br. J. Canc.* 86 (6) (2002) 983–988.
- [37] W.E. Hardman, M.P. Moyer, I.L. Cameron, Efficacy of treatment of colon, lung and breast human carcinoma xenografts with: doxorubicin, cisplatin, irinotecan or toptecan, *Anticancer Res.* 19 (3B) (1999) 2269–2274.
- [38] B. Xu, et al., Validated LC-MS/MS method for the determination of 3-hydroxyflavone and its glucuronide in blood and bioequivalent buffers: application to pharmacokinetic, absorption, and metabolism studies, *J. Pharmaceut. Biomed. Anal.* 85 (2013) 245–252.
- [39] E.D. Lephart, et al., Behavioral effects of endocrine-disrupting substances: phytoestrogens, *ILAR J.* 45 (4) (2004) 443–454.
- [40] P.G. Reeves, F.H. Nielsen, G.C. Fahey Jr., AIN-93 purified diets for laboratory rodents: final report of the American Institute of Nutrition ad hoc writing committee on the reformulation of the AIN-76A rodent diet, *J. Nutr.* 123 (11) (1993) 1939–1951.
- [41] J.W. Apolzan, R.B. Harris, Differential effects of chow and purified diet on the consumption of sucrose solution and lard and the development of obesity, *Physiol. Behav.* 105 (2) (2012) 325–331.
- [42] N.M. Brown, K.D. Setchell, Animal models impacted by phytoestrogens in commercial chow: implications for pathways influenced by hormones, *Lab. Invest.* 81 (5) (2001) 735–747.
- [43] J.E. Thigpen, et al., Phytoestrogen content of purified, open- and closed-formula laboratory animal diets, *Lab. Anim. Sci.* 49 (5) (1999) 530–536.
- [44] in Fat Content and Composition of Animal Products, Proceedings of a Symposium Washington, D.C. December 12–13, 1974, 1976 Washington (DC).
- [45] US National Nutrient Database, R, Soybean Oil, Salad or Cooking, Fat Composition, 100 g, United States Department of Agriculture, May 2016 Retrieved 6 September 2017.
- [46] US National Nutrient Database, R, Corn Oil, Industrial and Retail, All Purpose Salad or Cooking, Fat composition, 100 g, United States Department of Agriculture, May 2016 Retrieved 6 September 2017.
- [47] A. Kurita, et al., Modified irinotecan hydrochloride (CPT-11) administration schedule improves induction of delayed-onset diarrhea in rats, *Canc. Chemother. Pharmacol.* 46 (3) (2000) 211–220.
- [48] E.M. Brunt, et al., Nonalcoholic steatohepatitis: a proposal for grading and staging the histological lesions, *Am. J. Gastroenterol.* 94 (9) (1999) 2467–2474.
- [49] D.E. Kleiner, et al., Design and validation of a histological scoring system for nonalcoholic fatty liver disease, *Hepatology* 41 (6) (2005) 1313–1321.
- [50] S. Basu, et al., Development and validation of an UPLC-MS/MS method for the quantification of irinotecan, SN-38 and SN-38 glucuronide in plasma, urine, feces, liver and kidney: application to a pharmacokinetic study of irinotecan in rats, *J. Chromatogr. B Analyt. Technol. Biomed. Life Sci.* (2016) 1015–1016 34–41.
- [51] P. Mallick, et al., Role of Toll-like receptor 4 in drug-drug interaction between paclitaxel and irinotecan in vitro, *Toxicol. Vitro* 41 (2017) 75–82.
- [52] A. Rahman, et al., Selective biotransformation of taxol to 6 alpha-hydroxytaxol by human cytochrome P450 2C8, *Cancer Res.* 54 (21) (1994) 5543–5546.
- [53] P. Mallick, et al., Impact of obesity on accumulation of the toxic irinotecan metabolite, SN-38, in mice, *Life Sci.* 139 (2015) 132–138.
- [54] M.R. Brzezinski, et al., Purification and characterization of a human liver cocaine carboxylesterase that catalyzes the production of benzoylecgonine and the formation of cocaethylene from alcohol and cocaine, *Biochem. Pharmacol.* 48 (9) (1994) 1747–1755.
- [55] T. Marshall, P. Shult, W.W. Busse, Release of lysosomal enzyme beta-glucuronidase from isolated human eosinophils, *J. Allergy Clin. Immunol.* 82 (4) (1988) 550–555.
- [56] K. Shimoi, et al., Deglucuronidation of a flavonoid, luteolin monoglucuronide, during inflammation, *Drug Metab. Dispos.* 29 (12) (2001) 1521–1524.
- [57] K. Takasuna, et al., Involvement of beta-glucuronidase in intestinal microflora in the intestinal toxicity of the antitumor camptothecin derivative irinotecan hydrochloride (CPT-11) in rats, *Cancer Res.* 56 (16) (1996) 3752–3757.
- [58] K. Takasuna, et al., Optimal anti-diarrhea treatment for antitumor agent irinotecan hydrochloride (CPT-11)-induced delayed diarrhea, *Canc. Chemother. Pharmacol.* 58 (4) (2006) 494–503.
- [59] R.A. Darling, et al., Incubation of food craving is independent of macronutrient composition, *Sci. Rep.* 6 (2016) 30900.
- [60] I. Torre-Villalvazo, et al., Soy protein ameliorates metabolic abnormalities in liver and adipose tissue of rats fed a high fat diet, *J. Nutr.* 138 (3) (2008) 462–468.
- [61] E.D. Lephart, et al., Dietary isoflavones alter regulatory behaviors, metabolic hormones and neuroendocrine function in Long-Evans male rats, *Nutr. Metab.* 1 (1) (2004) 16.
- [62] J.R. Levy, J.N. Clore, W. Stevens, Dietary n-3 polyunsaturated fatty acids decrease hepatic triglycerides in Fischer 344 rats, *Hepatology* 39 (3) (2004) 608–616.
- [63] L.H. Storlien, et al., Fish oil prevents insulin resistance induced by high-fat feeding in rats, *Science* 237 (4817) (1987) 885–888.
- [64] A.M. Stephen, J.H. Cummings, Mechanism of action of dietary fibre in the human colon, *Nature* 284 (5753) (1980) 283–284.
- [65] J.H. Cummings, et al., Fecal weight, colon cancer risk, and dietary intake of non-starch polysaccharides (dietary fiber), *Gastroenterology* 103 (6) (1992) 1783–1789.
- [66] A.M. Birkett, et al., Dietary intake and faecal excretion of carbohydrate by Australians: importance of achieving stool weights greater than 150 g to improve faecal markers relevant to colon cancer risk, *Eur. J. Clin. Nutr.* 51 (9) (1997) 625–632.
- [67] R. Kubant, et al., A comparison of effects of lard and hydrogenated vegetable shortening on the development of high-fat diet-induced obesity in rats, *Nutr. Diabetes* 5 (2015) e188.

- [68] R. Buettner, et al., Defining high-fat-diet rat models: metabolic and molecular effects of different fat types, *J. Mol. Endocrinol.* 36 (3) (2006) 485–501.
- [69] K.L. Stanhope, P.J. Havel, Fructose consumption: potential mechanisms for its effects to increase visceral adiposity and induce dyslipidemia and insulin resistance, *Curr. Opin. Lipidol.* 19 (1) (2008) 16–24.
- [70] M.B. Vos, J.E. Lavine, Dietary fructose in nonalcoholic fatty liver disease, *Hepatology* 57 (6) (2013) 2525–2531.
- [71] H. Goto, et al., Effects of fructo-oligosaccharide on DSS-induced colitis differ in mice fed nonpurified and purified diets, *J. Nutr.* 140 (12) (2010) 2121–2127.
- [72] N.M. Moreau, et al., Restoration of the integrity of rat caeco-colonic mucosa by resistant starch, but not by fructo-oligosaccharides, in dextran sulfate sodium-induced experimental colitis, *Br. J. Nutr.* 90 (1) (2003) 75–85.
- [73] M.K. Danks, et al., Overexpression of a rabbit liver carboxylesterase sensitizes human tumor cells to CPT-11, *Cancer Res.* 58 (1) (1998) 20–22.
- [74] K. Sai, et al., Association of carboxylesterase 1A genotypes with irinotecan pharmacokinetics in Japanese cancer patients, *Br. J. Clin. Pharmacol.* 70 (2) (2010) 222–233.
- [75] R.H. Mathijssen, et al., Clinical pharmacokinetics and metabolism of irinotecan (CPT-11), *Clin. Canc. Res.* 7 (8) (2001) 2182–2194.
- [76] G.R. Wilkinson, The effects of diet, aging and disease-states on presystemic elimination and oral drug bioavailability in humans, *Adv. Drug Deliv. Rev.* 27 (2–3) (1997) 129–159.
- [77] X.B. Lin, et al., The role of intestinal microbiota in development of irinotecan toxicity and in toxicity reduction through dietary fibres in rats, *PLoS One* 9 (1) (2014) e83644.
- [78] Y. Ando, et al., Polymorphisms of UDP-glucuronosyltransferase gene and irinotecan toxicity: a pharmacogenetic analysis, *Cancer Res.* 60 (24) (2000) 6921–6926.
- [79] S. Chen, et al., Intestinal glucuronidation protects against chemotherapy-induced toxicity by irinotecan (CPT-11), *Proc. Natl. Acad. Sci. U. S. A.* 110 (47) (2013) 19143–19148.
- [80] M.L. Rothenberg, et al., A multicenter, phase II trial of weekly irinotecan (CPT-11) in patients with previously treated colorectal carcinoma, *Cancer* 85 (4) (1999) 786–795.
- [81] H.C. Pitot, et al., Phase II trial of irinotecan in patients with metastatic colorectal carcinoma, *J. Clin. Oncol.* 15 (8) (1997) 2910–2919.
- [82] A.A. Parikh, et al., Perioperative complications in patients undergoing major liver resection with or without neoadjuvant chemotherapy, *J. Gastrointest. Surg.* 7 (8) (2003) 1082–1088.
- [83] A. Sahajpal, et al., Chemotherapy for colorectal cancer prior to liver resection for colorectal cancer hepatic metastases does not adversely affect peri-operative outcomes, *J. Surg. Oncol.* 95 (1) (2007) 22–27.